

Vibration of Mechanical Systems: Principles and Practice

Vibration of 2DOF Systems: Forced Response of Damped Systems

Example: Double compound pendulum

Consider the double compound pendulum discussed in class. The system consists of two uniform rigid links of mass m_1 and m_2 , and length l_1 and l_2 , respectively.

- Derive the governing equations of motion by the Lagrange equations.
- Suppose the links are identical, obtain the natural frequencies and modes of vibration.
- Suppose the links are of the same density and denote the ratio of $l_2/l_1 = \alpha$. Examine the effects of α on the natural frequencies and modes of vibration for this system. Interpret your results.

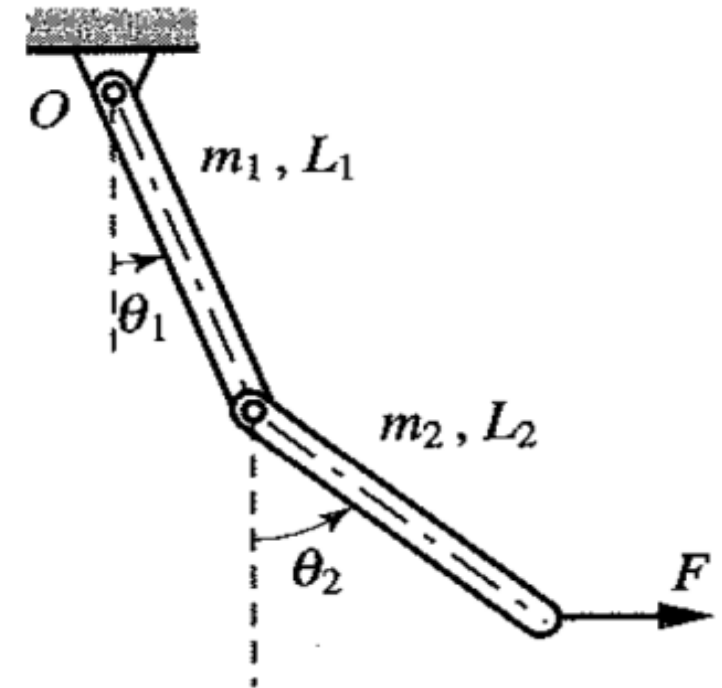
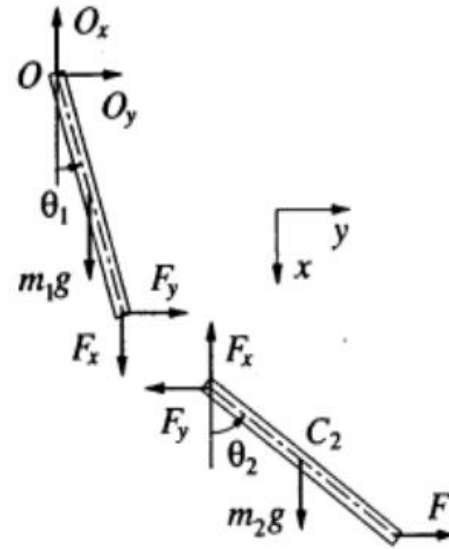


FIGURE 1.60
Two-link system

Equations of motion by Newtonian approach:



$$\Sigma M_O = I_O \ddot{\theta}_1 = -m_1 g \frac{L_1}{2} \sin \theta_1 + F_y L_1 \cos \theta_1 - F_x L_1 \sin \theta_1 \quad (a)$$

$$\Sigma F_{x_2} = m_2 a_{C_2x} = m_2 \left(L_1 \ddot{\theta}_1 \sin \theta_1 + L_1 \dot{\theta}_1^2 \cos \theta_1 + \frac{L_2}{2} \ddot{\theta}_2 \sin \theta_2 + \frac{L_2}{2} \dot{\theta}_2^2 \cos \theta_2 \right) = F_x - m_2 g \quad (b)$$

$$\Sigma F_{y_2} = m_2 a_{C_2y} = m_2 \left(L_1 \ddot{\theta}_1 \cos \theta_1 - L_1 \dot{\theta}_1^2 \sin \theta_1 + \frac{L_2}{2} \ddot{\theta}_2 \cos \theta_2 - \frac{L_2}{2} \dot{\theta}_2^2 \sin \theta_2 \right) = F - F_y \quad (c)$$

$$\Sigma M_{C_2} = I_{C_2} \ddot{\theta}_2 = F \frac{L_2}{2} \cos \theta_2 + F_y \frac{L_2}{2} \cos \theta_2 - F_x \frac{L_2}{2} \sin \theta_2 \quad (d)$$

$$\text{where } I_O = \frac{1}{3} m_1 L_1^2, \quad I_{C_2} = \frac{1}{12} m_2 L_2^2$$

From Eqs. (b) and (c),

$$F_{x_2} = m_2 \left(L_1 \ddot{\theta}_1 \sin \theta_1 + L_1 \dot{\theta}_1^2 \cos \theta_1 + \frac{L_2}{2} \ddot{\theta}_2 \sin \theta_2 + \frac{L_2}{2} \dot{\theta}_2^2 \cos \theta_2 \right) + m_2 g$$

$$F_{y_2} = F - m_2 \left(L_1 \ddot{\theta}_1 \cos \theta_1 - L_1 \dot{\theta}_1^2 \sin \theta_1 + \frac{L_2}{2} \ddot{\theta}_2 \cos \theta_2 - \frac{L_2}{2} \dot{\theta}_2^2 \sin \theta_2 \right)$$

Inserting F_{x_2} and F_{y_2} into Eqs. (a) and (d) and rearranging,

$$\left(\frac{m_1}{3} + m_2 \right) L_1 \ddot{\theta}_1 + \frac{m_2 L_2}{2} \cos(\theta_2 - \theta_1) \ddot{\theta}_2 - \frac{m_2 L_2}{2} \sin(\theta_2 - \theta_1) \dot{\theta}_2^2 + \left(\frac{m_1}{2} + m_2 \right) g \sin \theta_1 = F \cos \theta_1$$

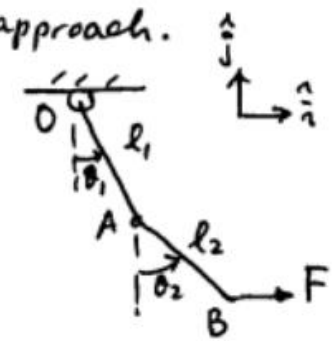
$$m_2 L_1 \cos(\theta_2 - \theta_1) \ddot{\theta}_1 + \frac{2}{3} m_2 L_2 \ddot{\theta}_2 + m_2 L_1 \sin(\theta_2 - \theta_1) \dot{\theta}_1^2 + m_2 g \sin \theta_2 = 2F \cos \theta_2$$

Equations of motion by Lagrangian approach:

(A) Derive the governing equations of motion by energy approach.
Employ the Lagrange's equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_j} \right) - \frac{\partial L}{\partial \theta_j} = Q_j, \quad j=1,2$$

(1)



where $L = T - V$,

θ_j = generalized coordinates

Q_j = generalized forces

$$T_1 = \frac{1}{6} m_1 l_1^2 \dot{\theta}_1^2, \quad T_2 = \frac{1}{24} m_2 l_2^2 \dot{\theta}_2^2 + \frac{1}{2} m_2 v_{G_2}^2$$

$$\underline{v}_{G_2} = (l_1 \dot{\theta}_1 \cos \theta_1 \hat{i} + l_1 \dot{\theta}_1 \sin \theta_1 \hat{j}) + \left(\frac{l_2}{2} \dot{\theta}_2 \cos \theta_2 \hat{i} + \frac{l_2}{2} \dot{\theta}_2 \sin \theta_2 \hat{j} \right)$$

$$v_{G_2}^2 = \underline{v}_{G_2} \cdot \underline{v}_{G_2} = l_1^2 \dot{\theta}_1^2 + \frac{l_2^2}{4} \dot{\theta}_2^2 + l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

$$\text{Thus, } T_2 = \frac{m_2}{2} l_1^2 \dot{\theta}_1^2 + \frac{1}{6} m_2 l_2^2 \dot{\theta}_2^2 + \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

The kinetic energy of the system is thus

$$T = T_1 + T_2 \Rightarrow T = \left(\frac{m_1}{6} l_1^2 + \frac{m_2}{2} l_1^2 \right) \dot{\theta}_1^2 + \frac{1}{6} m_2 l_2^2 \dot{\theta}_2^2 + \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

The potential energies of the bars are (setting datum at the pivot)

$$V_1 = -m_1 g \frac{l_1}{2} \cos \theta_1, \quad V_2 = -m_2 g (l_1 \cos \theta_1 + \frac{l_2}{2} \cos \theta_2)$$

The generalized forces are obtained from the virtual work.

$$\delta W_{nc} = \underline{F} \cdot \delta \underline{r}_B = \sum_{j=1}^2 Q_j \delta \theta_j$$

$$\begin{aligned} \underline{r}_B &= \underline{r}_A + \underline{r}_{B/A} \\ &= (l_1 \sin \theta_1 \hat{i} - l_1 \cos \theta_1 \hat{j}) \\ &\quad + (l_2 \sin \theta_2 \hat{i} - l_2 \cos \theta_2 \hat{j}) \\ \delta \underline{r}_B &= (l_1 \cos \theta_1 \delta \theta_1 \hat{i} + l_1 \sin \theta_1 \delta \theta_1 \hat{j}) \\ &\quad + \dots \end{aligned}$$

$$\Rightarrow \delta W_{nc} = F \hat{i} \cdot \delta [(l_1 \sin \theta_1 + l_2 \sin \theta_2) \hat{i} + (-l_1 \cos \theta_1 - l_2 \cos \theta_2) \hat{j}]$$

$$= \underbrace{F l_1 \cos \theta_1}_{Q_1} \delta \theta_1 + \underbrace{F l_2 \cos \theta_2}_{Q_2} \delta \theta_2$$

$$\frac{\partial L}{\partial \dot{\theta}_1} = \left(\frac{m_1 l_1^2}{3} + m_2 l_1^2 \right) \dot{\theta}_1 + \frac{m_2}{2} l_1 l_2 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = \frac{m_2 l_2^2}{3} \dot{\theta}_2 + \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \cos(\theta_2 - \theta_1)$$

$$-\frac{\partial L}{\partial \theta_1} = \left(\frac{m_1 g l_1}{2} + m_2 g l_1 \right) \sin \theta_1 - \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1)$$

$$-\frac{\partial L}{\partial \theta_2} = \frac{m_2 g l_2}{2} \sin \theta_2 + \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = \left(\frac{m_1 l_1^2}{3} + m_2 l_1^2 \right) \ddot{\theta}_1 + \frac{m_2}{2} l_1 l_2 \ddot{\theta}_2 \cos(\theta_2 - \theta_1)$$

$$- \frac{m_2}{2} l_1 l_2 \dot{\theta}_2 \sin(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = \frac{m_2 l_2^2}{3} \ddot{\theta}_2 + \frac{m_2}{2} l_1 l_2 \ddot{\theta}_1 \cos(\theta_2 - \theta_1) - \frac{m_2}{2} l_1 l_2 \dot{\theta}_1 \sin(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

Substituting above terms into (1) gives the equations of motion,

$$\left(\frac{m_1}{3} + m_2 \right) l_1 \ddot{\theta}_1 + \frac{m_2 l_2}{2} \cos(\theta_2 - \theta_1) \ddot{\theta}_2 - \frac{m_2 l_2}{2} \sin(\theta_2 - \theta_1) \dot{\theta}_2^2$$

$$+ \left(\frac{m_1}{2} + m_2 \right) g \sin \theta_1 = F \cos \theta_1 \quad (2)$$

$$\frac{2}{3} m_2 l_2 \ddot{\theta}_2 + m_2 l_1 \ddot{\theta}_1 \cos(\theta_2 - \theta_1) + m_2 l_1 \dot{\theta}_1^2 \sin(\theta_2 - \theta_1)$$

$$+ m_2 g \sin \theta_2 = 2 F \cos \theta_2 \quad (3)$$

//

(B) An alternative approach is to apply the work-energy principle.

$$\Delta T + \Delta V = W_{nc} \Rightarrow \frac{d}{dt}(T+V) = \dot{W}_{nc} \quad (4)$$

$$\text{where } \frac{dT}{dt} = \frac{\partial T}{\partial \dot{\theta}_1} \dot{\theta}_1 + \frac{\partial T}{\partial \dot{\theta}_2} \dot{\theta}_2 + \frac{\partial T}{\partial \ddot{\theta}_1} \ddot{\theta}_1 + \frac{\partial T}{\partial \ddot{\theta}_2} \ddot{\theta}_2 + \cancel{\frac{\partial T}{\partial t}}^0$$

$$\frac{dV}{dt} = \frac{\partial V}{\partial \dot{\theta}_1} \dot{\theta}_1 + \frac{\partial V}{\partial \dot{\theta}_2} \dot{\theta}_2 + \cancel{\frac{\partial V}{\partial t}}^0$$

$$\dot{W}_{nc} = \int \mathbf{F} \cdot \frac{d\mathbf{r}_B}{dt} \quad (\text{since } \mathbf{F} \text{ is time-invariant})$$

$$= F l_1 \cos \theta_1 \dot{\theta}_1 + F l_2 \cos \theta_2 \dot{\theta}_2$$

Expanding the terms above and substituting into (4) gives

$$(\dots) \dot{\theta}_1 + (\dots) \dot{\theta}_2 = 0 \quad (5)$$

By setting each of the two brackets in (5) to zero, we obtain the governing equations of motion (3) and (4), respectively.

Note that the main advantage of energy-based methods is that we do not have to deal with the constraint forces (action and reaction forces in most cases). The algebra is thus greatly simplified.

(b) The linearized governing equations of motion about the trivial equilibrium are:

$$\begin{aligned} \left(\frac{m_1}{3} + m_2\right)\ell_1\ddot{\theta}_1 + \frac{m_2\ell_2}{2}\ddot{\theta}_2 + \left(\frac{m_1}{2} + m_2\right)g\theta_1 &= F \\ m_2\ell_1\ddot{\theta}_1 + \frac{2m_2\ell_2}{3}\ddot{\theta}_2 + m_2g\theta_2 &= 2F \end{aligned} \quad (6)$$

For identical links, $m_1 = m_2$ and $\ell_1 = \ell_2$. To solve for the natural frequencies (eigenvalues) and modes of vibration (eigenvectors), consider the free vibration problem. The governing equations simplify to:

$$\begin{bmatrix} 4/3 & 1/2 \\ 1 & 2/3 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \frac{g}{\ell} \begin{bmatrix} 3/2 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0 \quad (7)$$

Note that the degrees of freedom are coupled only through the inertia (the mass matrix). Let the general solution be of the form $\mathbf{q} = \mathbf{u}e^{i\omega t}$, the eigenvalue problem is

$$\mathbf{K}\mathbf{u} = \lambda\mathbf{M}\mathbf{u}$$

where $\lambda = \frac{\omega^2\ell}{g}$ is the normalized eigenvalue and $\mathbf{M} = \begin{bmatrix} 4/3 & 1/2 \\ 1 & 2/3 \end{bmatrix}$ and

$$\mathbf{K} = \begin{bmatrix} 3/2 & 0 \\ 0 & 1 \end{bmatrix}.$$

MATLAB calculations using eig:

```
>> M=[4/3 1/2; 1 2/3]; K = diag([3/2 1]);
>> [V D] = eig(K,M)
```

V =

0.6991 -0.4768
1.0000 1.0000

D =

0.7322 0
0 5.2678

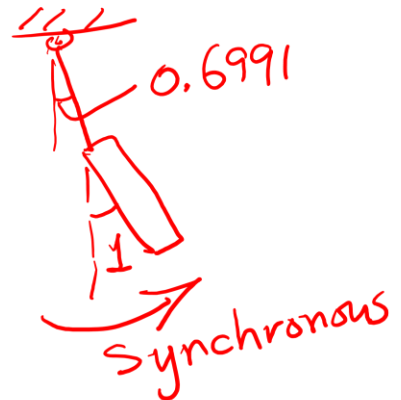
$$0.7322 = \omega_1^2 \frac{\ell}{g}$$

$$\omega_1 = \sqrt{\frac{0.7322g}{\ell}}$$

asynchronous

first mode

2nd mode



$\ell = \dots$



MATLAB results:

```
>> M=[4/3 1/2; 1 2/3]; K = diag([3/2 1]);  
>> [V D] = eig(K,M)
```

V =

```
0.6991    -0.4768  
1.0000     1.0000
```

Related to natural modes or normal modes

D =

```
0.7322     0  
0     5.2678
```

Related to eigenvalues and natural frequencies

The frequency equation is $\det(\mathbf{K} - \tilde{\lambda}\mathbf{M}) = 0$, where $\mathbf{K}$ and $\mathbf{M}$ are given by the numerical values shown before, from which the natural frequencies can be found. Solving the frequency equation gives $\tilde{\lambda} = 0.7322, 5.2678$. The square roots of $\tilde{\lambda}$ are 0.8556 and 2.2951.

$$\text{Thus } \omega_1 = 0.8556 \sqrt{\frac{g}{L}}, \omega_2 = 2.2951 \sqrt{\frac{g}{L}}.$$

$\tilde{\lambda}_1 = 0.7322$ case: Solving $\mathbf{K}\mathbf{u}_1 = \lambda\mathbf{M}\mathbf{u}_1$ gives $\mathbf{u}_1 = \alpha_1 \begin{Bmatrix} 0.6991 \\ 1 \end{Bmatrix}$, where α_1 is the normalization constant.

Applying normalization $\mathbf{u}_1^T \mathbf{M} \mathbf{u}_1 = 1$ gives $\mathbf{u}_1 = 0.65 \begin{Bmatrix} 0.6991 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 0.454 \\ 0.65 \end{Bmatrix}$.

first normal mode

$\tilde{\lambda}_2 = 5.2678$ case: Solving $\mathbf{K}\mathbf{u}_2 = \lambda\mathbf{M}\mathbf{u}_2$ gives $\mathbf{u}_2 = \alpha_2 \begin{Bmatrix} -0.4768 \\ 1 \end{Bmatrix}$, where α_2 is the normalization constant.

Applying normalization $\mathbf{u}_2^T \mathbf{M} \mathbf{u}_2 = 1$ gives $\mathbf{u}_2 = 1.982 \begin{Bmatrix} -0.4768 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -0.945 \\ 1.982 \end{Bmatrix}$.

**second normal mode
(there is a nodal point)**

Thus, the modal matrix is

$$\mathbf{U} = \begin{bmatrix} 0.454 & -0.945 \\ 0.65 & 1.982 \end{bmatrix}$$

Modal matrix

The general response is given by the modal expansion:

$$\mathbf{x} = \sum_{k=1}^2 q_k(t) \mathbf{u}_k, \text{ or } \mathbf{x} = \mathbf{U}\mathbf{q}, \text{ where } \mathbf{x} = \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} \text{ and } \mathbf{q} = \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix}$$

where, the generalized coordinate $q_k(t)$ satisfies the differential equation

$$\ddot{q}_k + \omega_k^2 q_k = 0, \quad \mathbf{q}(0) = \mathbf{U}^T \mathbf{M} \mathbf{x}(0), \dot{\mathbf{q}}(0) = \mathbf{U}^T \mathbf{M} \dot{\mathbf{x}}(0)$$

The solution for the generalized coordinates $q_k(t)$ is

$$q_k(t) = q_{k0} \cos(\omega_k t) + \frac{\dot{q}_{k0}}{\omega_k} \sin(\omega_k t), \quad k = 1, 2$$

$$\theta_1(0) = 2^\circ, \dot{\theta}_1(0) = 0, \theta_2(0) = 0, \dot{\theta}_2(0) = 0$$

Denote $\theta_0 = 2^\circ = \frac{2\pi}{180}$ rad (initial angle of the left pendulum). For this problem, we have

$$\mathbf{x}(0) = \mathbf{x}_0 = \begin{Bmatrix} \theta_0 \\ 0 \end{Bmatrix}, \quad \dot{\mathbf{x}}(0) = 0.$$

Hence,

$$\mathbf{q}(0) = \theta_0 \begin{Bmatrix} 1.255 \\ 0.722 \end{Bmatrix}, \quad \dot{\mathbf{q}}(0) = 0$$

The free response is then given by

$$\mathbf{x} = \begin{Bmatrix} \theta_1(t) \\ \theta_2(t) \end{Bmatrix} = \sum_{k=1}^2 q_k(t) \mathbf{u}_k = 1.255\theta_0 \cos(\omega_1 t) \begin{Bmatrix} 0.454 \\ 0.65 \end{Bmatrix} + 0.722\theta_0 \cos(\omega_2 t) \begin{Bmatrix} -0.945 \\ 1.982 \end{Bmatrix}$$

where, the natural frequencies for this problem are $\omega_1 = 0.8556 \sqrt{\frac{g}{L}}$, $\omega_2 = 2.2951 \sqrt{\frac{g}{L}}$.

Forced vibration analysis (undamped systems):

For undamped systems, when forces are applied,

$$\ddot{q}_k + \omega_k^2 q_k = Q_k, \quad \mathbf{q}(0) = \mathbf{U}^T \mathbf{M} \mathbf{x}(0), \dot{\mathbf{q}}(0) = \mathbf{U}^T \mathbf{M} \dot{\mathbf{x}}(0), \mathbf{Q} = \mathbf{U}^T \mathbf{F}$$

The solution for the generalized coordinates $q_k(t)$ is

$$q_k(t) = q_{k0} \cos(\omega_k t) + \frac{\dot{q}_{k0}}{\omega_k} \sin(\omega_k t) + \frac{1}{\omega_k} \int_0^t Q_k(\tau) \sin \omega_k(t - \tau) d\tau, \quad k = 1, 2$$

The key to solve for the vibration response of more than one-degree-of-freedom systems is to **decouple** the differential equations into SDOF oscillators by the modal analysis technique. Then apply equations for SDOF systems to obtain the free and forced responses.

c) Denote $m_1 = m$, $m_2 = \alpha m$, and $\ell_1 = \ell$, $\ell_2 = \alpha \ell$. To solve for the natural frequencies (eigenvalues) and modes of vibration (eigenvectors), consider the free vibration problem. The governing equations simplify to:

$$\begin{bmatrix} 1/3 + \alpha & \alpha^2/2 \\ \alpha & 2\alpha^2/3 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \frac{g}{\ell} \begin{bmatrix} 1/2 + \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0 \quad (8)$$

```
>> for a = [0.1 1 10] % obtain results for alpha = 0.1,
    1, 10
    fprintf('alpha = %6.2f', a)
    M = [1/3+a a^2/2; a 2*a^2/3]; K = diag([1/2+a a]);
    [V D] = eig(K,M)
end
alpha =    0.10
V =
```

```
    0.6680    -0.0125
    1.0000     1.0000
```

```
D =
```

```
    1.3611         0
         0    18.4528
```

$\alpha = 1.00$

$V =$

0.6991	-0.4768
1.0000	1.0000

$D =$

0.7322	0
0	5.2678

$\alpha = 10.00$

$V =$

1.0000	0.7413
-0.1557	1.0000

$D =$

4.1180	0
0	0.1350

Note that as α increases, the frequency associated with the lower link reduces and there is a switch of mode shapes (highlighted).

Example: A vehicle bounce-pitch model

Determine the pitch (angular motion) and bounce (up-and-down linear motion) frequencies and the location of oscillation centers (nodes) of an automobile with the following data (see Fig. 5.13):

Mass (m) = 1000 kg

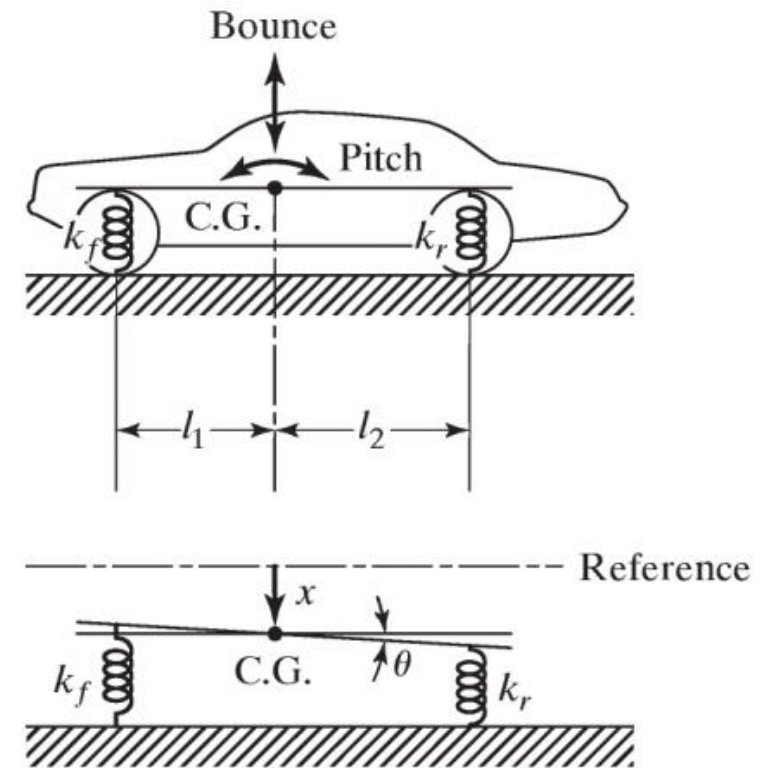
Radius of gyration (r) = 0.9 m

Distance between front axle and C.G. (l_1) = 1.0 m

Distance between rear axle and C.G. (l_2) = 1.5 m

Front spring stiffness (k_f) = 18 kN/m

Rear spring stiffness (k_r) = 22 kN/m



Solution: If x and θ are used as independent coordinates, the equations of motion are given by Eq. (5.23) with $k_1 = k_f$, $k_2 = k_r$, and $J_0 = mr^2$. For free vibration, we assume a harmonic solution:

$$x(t) = X \cos(\omega t + \phi), \quad \theta(t) = \Theta \cos(\omega t + \phi) \quad (\text{E.1})$$

Using Eqs. (E.1) and (5.23), we obtain

$$\begin{bmatrix} (-m\omega^2 + k_f + k_r) & (-k_f l_1 + k_r l_2) \\ (-k_f l_1 + k_r l_2) & (-J_0\omega^2 + k_f l_1^2 + k_r l_2^2) \end{bmatrix} \begin{Bmatrix} X \\ \Theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (\text{E.2})$$

For the known data, Eq. (E.2) becomes

$$\begin{bmatrix} (-1000\omega^2 + 40,000) & 15,000 \\ 15,000 & (-810\omega^2 + 67,500) \end{bmatrix} \begin{Bmatrix} X \\ \Theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (\text{E.3})$$

from which the frequency equation can be derived:

$$8.1\omega^4 - 999\omega^2 + 24,750 = 0 \quad (\text{E.4})$$

The natural frequencies can be found from Eq. (E.4):

$$\omega_1 = 5.8593 \text{ rad/s}, \quad \omega_2 = 9.4341 \text{ rad/s} \quad (\text{E.5})$$

With these values, the ratio of amplitudes can be found from Eq. (E.3):

$$\frac{X^{(1)}}{\Theta^{(1)}} = -2.6461, \quad \frac{X^{(2)}}{\Theta^{(2)}} = 0.3061 \quad (\text{E.6})$$

The node locations can be obtained by noting that the tangent of a small angle is approximately equal to the angle itself. Thus, from Fig. 5.14, we find that the distance between the C.G. and the node is -2.6461 m for ω_1 and 0.3061 m for ω_2 . The mode shapes are shown by dashed lines in Fig. 5.14.

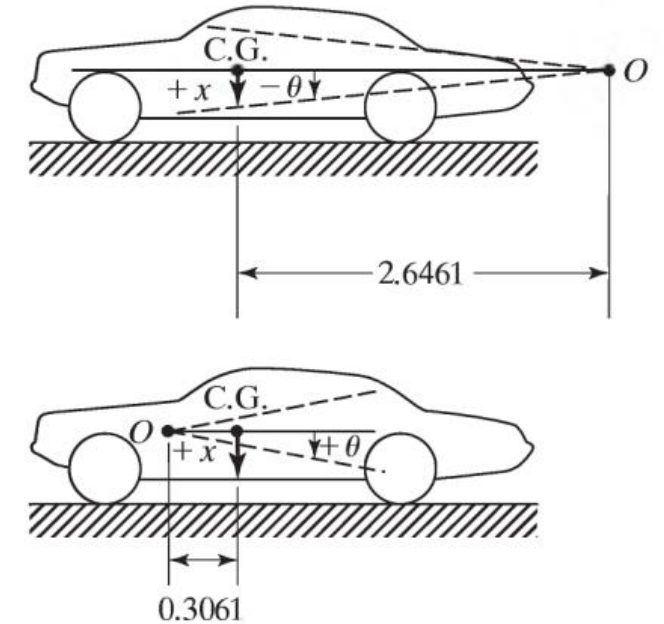


FIGURE 5.14 Mode shapes of an automobile.

Decoupling of the forced vibration of 2DOF damped systems:

To evaluate the forced vibration response (due to excitations), consider

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}(t), \quad \mathbf{x}(0) = \begin{Bmatrix} x_{10} \\ x_{20} \end{Bmatrix}, \quad \dot{\mathbf{x}}(0) = \begin{Bmatrix} v_{10} \\ v_{20} \end{Bmatrix}$$

Here, we assume **proportional damping**, $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$. This is a sufficient condition to decouple the system into its normal modes and allow us to write the response in the general form as:

$$\mathbf{x}(t) = \sum_{k=1}^n q_k(t) \mathbf{u}_k = \mathbf{U}\mathbf{q}, \quad \mathbf{U} = \text{modal matrix}, n = 2 \text{ for 2DOF systems}$$

where, q_k is the generalized or modal coordinate. The $\mathbf{u}_k$ is the eigenvector or modal vector, and n is the number of modes corresponding to the number of freedom of the vibratory system. Then, obtain

$$\begin{aligned} \mathbf{M}\mathbf{U}\ddot{\mathbf{q}} + \mathbf{C}\mathbf{U}\dot{\mathbf{q}} + \mathbf{K}\mathbf{U}\mathbf{q} &= \mathbf{F} \\ \mathbf{U}^T(\mathbf{M}\mathbf{U}\ddot{\mathbf{q}} + \mathbf{C}\mathbf{U}\dot{\mathbf{q}} + \mathbf{K}\mathbf{U}\mathbf{q}) &= \mathbf{U}^T\mathbf{F} \\ \ddot{\mathbf{q}} + (\alpha\mathbf{I} + \beta\mathbf{\Lambda})\dot{\mathbf{q}} + \mathbf{\Lambda}\mathbf{q} &= \mathbf{U}^T\mathbf{F} = \mathbf{Q} \end{aligned}$$

where, $\mathbf{Q}$ is the generalized force vector and $\mathbf{\Lambda}$ is the matrix of the eigenvalues

$$\mathbf{\Lambda} = \begin{bmatrix} \omega_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \omega_n^2 \end{bmatrix}, \quad \lambda_k = \omega_k^2$$

Define $\alpha + \beta\omega_k^2 = 2\zeta_k\omega_k$. Note that in the space of the modal coordinates, the system is decoupled into n SDOF oscillators. The total response for each generalised coordinate is simply the sum of the free response and the forced response. The decoupled equation for generalised coordinate $q_k(t)$ is:

$$\ddot{q}_k(t) + 2\zeta_k\omega_k\dot{q}_k(t) + \omega_k^2q_k(t) = Q_k, \quad k = 1, 2, \dots, n$$

$$q_k(t) = e^{-\zeta_k\omega_k t} q_k(0) \left(\cos \omega_{dk} t + \frac{\zeta_k}{\sqrt{1 - \zeta_k^2}} \sin \omega_{dk} t \right) + \frac{\dot{q}_k(0)}{\omega_{dk}} e^{-\zeta_k\omega_k t} \sin \omega_{dk} t$$

$$+ \underbrace{\frac{1}{\omega_{dk}} \int_0^t Q_k(\tau) e^{-\zeta_k\omega_k(t-\tau)} \sin \omega_{dk}(t-\tau) d\tau}_{\text{Forced response due to a general force by convolution integral with the impulse response function of SDOF system}}, k = 1, 2, \dots, n$$

where, $\omega_{dk} = \omega_k \sqrt{1 - \zeta_k^2}$, $q_k(0) = \mathbf{U}^{-1}x_k(0)$, $\dot{q}_k(0) = \mathbf{U}^{-1}\dot{x}_k(0)$, where numerically the inverse of $\mathbf{U}$ should be computed using the normalization condition $\mathbf{U}^T \mathbf{M} \mathbf{U} = \mathbf{I}$, thus $\mathbf{U}^{-1} = \mathbf{U}^T \mathbf{M}$. For harmonic excitations, the particular solution can be expressed in terms of the transfer functions.

Recall: Forced response to harmonic excitations (transfer function) for SDOF systems:

$$m\ddot{x}(t) + c\dot{x} + kx = F(t) = kA\cos\omega t \Rightarrow \ddot{x}(t) + 2\zeta\omega_n\dot{x} + \omega_n^2 x = A\omega_n^2 \cos\omega t$$

$$x(t) = X\cos(\omega t - \phi), \quad X = \frac{A}{\left\{ \left[1 - (\omega/\omega_n)^2 \right]^2 + (2\zeta\omega/\omega_n)^2 \right\}^{\frac{1}{2}}}, \quad \phi = \tan^{-1} \left(\frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2} \right)$$

Complex notations:

$$F(t) = kAe^{i\omega t} \quad X(i\omega) = AG(i\omega),$$

$$x(t) = AG(i\omega)e^{i\omega t}$$

$$G(i\omega) = \frac{1}{1 - (\omega/\omega_n)^2 + 2i\zeta\omega/\omega_n}$$

Notes:

- 1) $G(i\omega)$ is the frequency response function (FRF) or the system transfer function.
- 2) $G(i\omega)$ defines the input (force) and output (response) relationship.
- 3) Response amplitude: $X = A|G(i\omega)|$. $G(i\omega)$ also contains the phase information.
- 4) The poles (singularities) of $G(i\omega)$ are the eigenvalues of the system. Physically, these correspond to the resonance or the peak response amplitudes.

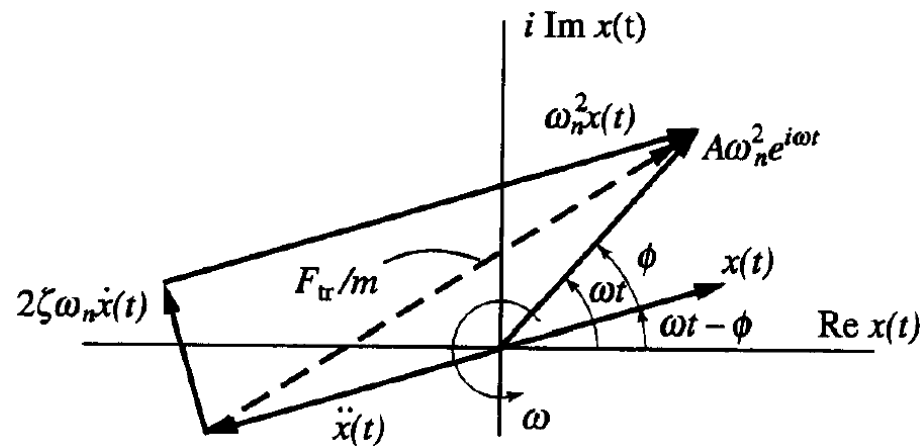
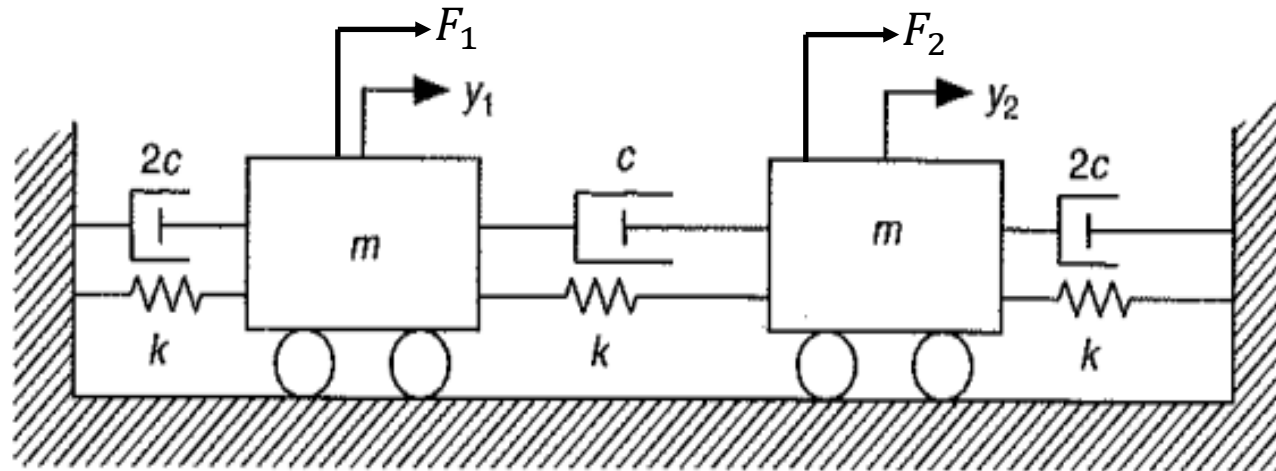


Fig. 2.5 Illustration of phase difference.

- Example:** (a) Derive the governing equation of the system.
- (b) Find the steady response of the system, $k = 1,000 \text{ N/m}$, $m = 500 \text{ kg}$,
 $F_1 = 500 \cos 10t$, $F_2 = 0$, $c = \alpha \mathbf{M} + \beta \mathbf{K}$, $\alpha = 1$, $\beta = 0.5$.
- (c) Suppose all initial conditions are zero, use MATLAB to obtain the steady response of the system, and compare the results with step (b).

Note that this system is proportionally damped and hence the governing equations can be decoupled.



(a) Using Newton's law, we can easily obtain the governing equations in the form of matrix:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} 3c & -c \\ -c & 3c \end{bmatrix} \begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} + \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

Since $c = \alpha \mathbf{M} + \beta \mathbf{K}$, thus $\alpha = c/m$ and $\beta = c/k$. Given $\alpha = 1, \beta = 0.5$, thus $c = 500 \text{ N-s/m}$.

(b) Assuming harmonic solution $y_i(t) = Y_i \cos \omega t$ for the free vibration problem, the equation for the eigenvalue problem is then:

$$\begin{bmatrix} -m\omega^2 + 2k & -k \\ -k & -m\omega^2 + 2k \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} -500\omega^2 + 2000 & -1000 \\ -1000 & -500\omega^2 + 2000 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (1)$$

Setting the determinant of the above to zero to find the frequency equation

$$\omega^4 - 8\omega^2 + 12 = 0$$

The solution gives the two natural frequencies

$$\omega_1 = \sqrt{2} \text{ rad/s}, \quad \omega_2 = \sqrt{6} \text{ rad/s}$$

Note that: $2\zeta_1\omega_1 = \alpha + \beta\omega_1^2 \Rightarrow \zeta_1 = 1/\sqrt{2}$; $2\zeta_2\omega_2 = \alpha + \beta\omega_2^2 \Rightarrow \zeta_2 = 2/\sqrt{6}$.

Alternate formulation of the eigenvalue problem:

Note that the frequency equation can also be solved by expressing the eigenvalues in terms of the basic frequency parameter $\sqrt{k/m}$. Define a non-dimensional eigenvalue $\lambda = \omega^2 m/k$. Thus, the eigenvalue problem $\mathbf{K}\mathbf{u} = \omega^2 \mathbf{M}\mathbf{u}$ becomes $\tilde{\mathbf{K}}\mathbf{u} = \lambda \tilde{\mathbf{M}}\mathbf{u}$, where

$$\tilde{\mathbf{K}} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad \tilde{\mathbf{M}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence, the frequency equation is obtained from solving

$$\det(\tilde{\mathbf{K}} - \lambda \tilde{\mathbf{M}}) = 0 \Rightarrow \begin{vmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{vmatrix} = 0$$

which gives the frequency equation

$$\lambda^2 - 4\lambda + 3 = 0$$

For this problem, we have defined $\lambda = \omega^2 m/k$. Hence, $\omega = \sqrt{\lambda} \sqrt{k/m} = \sqrt{\lambda} \sqrt{2}$. The solution to the frequency equation gives the two natural frequencies

$$\lambda_1 = 1 \Rightarrow \omega_1 = \sqrt{2} \text{ rad/s}, \quad \lambda_2 = 3 \Rightarrow \omega_2 = \sqrt{6} \text{ rad/s}$$

These are the same results as the previous calculations.

Substituting $\omega_1 = \sqrt{2}$, $\omega_2 = \sqrt{6}$ into the Equation (1), respectively, we can obtain the two eigenvectors (modal vectors). For this problem, the eigenvectors can also be obtained by applying the principle of symmetry

$$\mathbf{u}^{(1)} = \begin{bmatrix} u_1^{(1)} \\ u_2^{(1)} \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}^{(2)} = \begin{bmatrix} u_1^{(2)} \\ u_2^{(2)} \end{bmatrix} = \alpha_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Normalise the modal matrix with respect to the mass matrix, $\mathbf{U}^T \mathbf{M} \mathbf{U} = \mathbf{I}$,

$$\mathbf{U} = \frac{1}{\sqrt{2m}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Determine the generalised force vector

$$\mathbf{Q}(t) = \mathbf{U}^T \mathbf{F}(t) = \frac{1}{\sqrt{2m}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} F_1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2m}} \begin{bmatrix} F_1 \\ F_1 \end{bmatrix} = \begin{bmatrix} Q_{10} \\ Q_{20} \end{bmatrix} \cos 10t$$

where, $Q_{10} = \frac{500}{\sqrt{2m}} = 5\sqrt{10}$, $Q_{20} = \frac{500}{\sqrt{2m}} = 5\sqrt{10}$. Note that the unit for the Q is $\text{N}/\sqrt{\text{kg}}$. Finally, the system equations are **decoupled** into two SDOF oscillators in the modal space:

$$\ddot{q}_i(t) + 2\xi_i \omega_i \dot{q}_i(t) + \omega_i^2 q_i(t) = Q_i(t), \quad i = 1, 2 \quad (2)$$

Since the free vibration dissipates under damping (note that the homogeneous solution is non-zero under harmonic excitation), the steady-state solution of equation (2) is

$$q_i(t) = q_{i0} \cos(\omega t - \phi_i) \quad , i = 1, 2$$

where,

$$q_{i0} = \frac{Q_{i0}}{\omega_i^2} \frac{1}{\sqrt{(1 - r_i^2)^2 + (2\zeta_i r_i)^2}} \quad , \quad r_i = \omega / \omega_i \quad , \quad \omega = 10 \text{ rad/s}$$

$$\phi_i = \arctan\left(\frac{2\zeta_i r_i}{1 - r_i^2}\right)$$

Substituting the frequencies and generalized force into the above equation, we obtain

$$q_{10} = 0.1581, \quad q_{20} = 0.1548$$

$$\phi_1 = \arctan(10/-49) = 2.9403 \text{ rad (2nd quadrant)}$$

$$\phi_2 = \arctan(6.6667/-15.6667) = 2.7393 \text{ rad (2nd quadrant)}$$

Finally, we obtain the steady-state response of the system in physical coordinates,

$$\mathbf{y}(t) = \mathbf{U}\mathbf{q}(t) = \frac{1}{\sqrt{2m}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} = \frac{1}{\sqrt{2m}} \begin{bmatrix} q_1(t) + q_2(t) \\ q_1(t) - q_2(t) \end{bmatrix}$$

- (c) Solve by MATLAB ode solver. Rewrite the original equations in the state-space. Define the state-space variables as: $w_1 = y_1, w_2 = \dot{y}_1, w_3 = y_2, w_4 = \dot{y}_2$. The governing equations in physical coordinates are:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} 3c & -c \\ -c & 3c \end{bmatrix} \begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} + \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

The state-space equations are thus:

$$\frac{d\mathbf{w}}{dt} = \left\{ \begin{array}{l} \dot{y}_1 = w_2 \\ \ddot{y}_1 = \frac{500 \cos 10t + kw_3 - 2kw_1 + cw_4 - 3cw_2}{m} \\ \dot{y}_2 = w_4 \\ \ddot{y}_2 = \frac{-2kw_3 + kw_1 - 3cw_4 + cw_2}{m} \end{array} \right\}$$

Better (more systematic) Formulation of the State-Space Equations:

Note that the above formulation of the state-space equations has challenges if the mass matrix is not diagonal or easily invertible to allow the state-space equations to be written out explicitly. A better formulation is shown as follows in which the state-space equations can be expressed explicitly. Consider the system $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}(t)$ and define the state-space vector $\mathbf{w}$ as follows:

$$\mathbf{w} = \begin{Bmatrix} x_1(t) \\ \vdots \\ x_n(t) \\ \dot{x}_1(t) \\ \vdots \\ \dot{x}_n(t) \end{Bmatrix} = \begin{Bmatrix} \mathbf{x} \\ \dot{\mathbf{x}} \end{Bmatrix}$$

Thus,

$$\frac{d\mathbf{w}}{dt} = \begin{Bmatrix} \dot{\mathbf{x}} \\ \ddot{\mathbf{x}} \end{Bmatrix} = \begin{bmatrix} 0 & \mathbf{I}_n \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{Bmatrix} \mathbf{x} \\ \dot{\mathbf{x}} \end{Bmatrix} + \begin{Bmatrix} 0 \\ \mathbf{M}^{-1}\mathbf{F} \end{Bmatrix} = [\mathbf{A}]\mathbf{w} + \begin{Bmatrix} 0 \\ \mathbf{M}^{-1}\mathbf{F} \end{Bmatrix}$$

where, $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{A}$ is the state matrix. The above formulation can easily be coded for numerical computations. Just define the mass matrix and perform the inverse of the mass matrix numerically.

```
%% ZJU Vibration example, Forced vibration of 2DOF mass-spring-damper system
% Coded by Professor Chin-An Tan, 21 July 2021
% A 2DOF mass-spring-damper system is subjected to forces:
%  $F_1 = 500\cos(10t)$ ,  $F_2 = 0$ 
```

```
global m c k F0 omegaf
global invM C K
% Given data in the problem:
m = 500;      % mass in [kg]
k = 1e3;      % stiffness in [N/m]
alpha = 1;    % constant in proportional damping model
beta = 0.5;   % constant in proportional damping model
F0 = 500;     % amplitude of external force in [N]
omegaf = 10;  % frequency of external force in [rad/s]
% Constructed data
M = diag([m m]);
K = [2*k -k; -k 2*k];
C = alpha*M + beta*K;
invM = inv(M);
c = 500;      % damping in [N-s/m] (can be calculated; refer to notes)
% alternatively, C = [3*c -2*c; -2*c 3*c];
```

```

%% Eigenvalue problem and calculations
[V, D] = eig(K,M);
U = [1 1; 1 -1]/sqrt(2*m); % modal matrix
U.'*M*U % check whether this is identity matrix
U.'*C*U % check whether this is diagonal
U.'*K*U % check whether is the diagonal eigenvalue matrix D

CC = U.'*C*U; % damping matrix in the modal space
omegam = sqrt(D); % diagonal matrix of the natural frequencies
omega = diag(omegam); % a column vector of the natural frequencies
zeta = diag(CC)./omega/2; % damping ratio of the modal coordinates
% zeta(1) = damping ratio of first modal coordinate
% zeta(2) = damping ratio of second modal coordinate

```

```

%% Numerical solution by ode solver
% The state vector is defined w = [y1 y1dot y2 y2dot]
t0 = 0; tf = 5; % initial and final times of simulation
tspan = [t0, tf]; % duration of numerical simulation
w0 = [0 0 0 0]; % zero initial conditions
[t, w] = ode45(@dwdt,tspan,w0);

```

```

figure
subplot(2,1,1)
plot(t,w(:,1),'Linewidth',1.5), grid,
xlabel('t [sec]'), ylabel('y_1 [m]')
title('Response of the left mass m_1')
subplot(2,1,2)
plot(t,w(:,3),'r','Linewidth',1.5), grid,
xlabel('t [sec]'), ylabel('y_2 [m]')
title('Response of the right mass m_2')

```

← plot(t, w(:,2)) if using the alternate formulation of $\vec{w} = \{y_1, y_2, \dot{y}_1, \dot{y}_2\}^T$

```

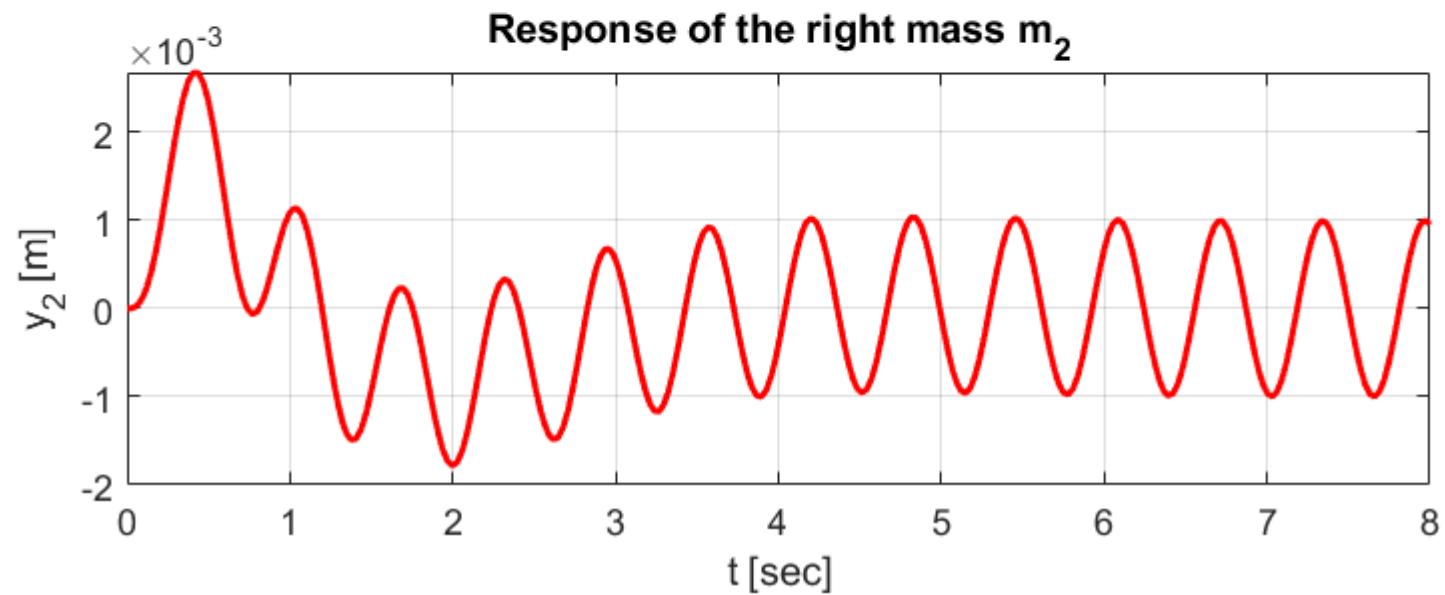
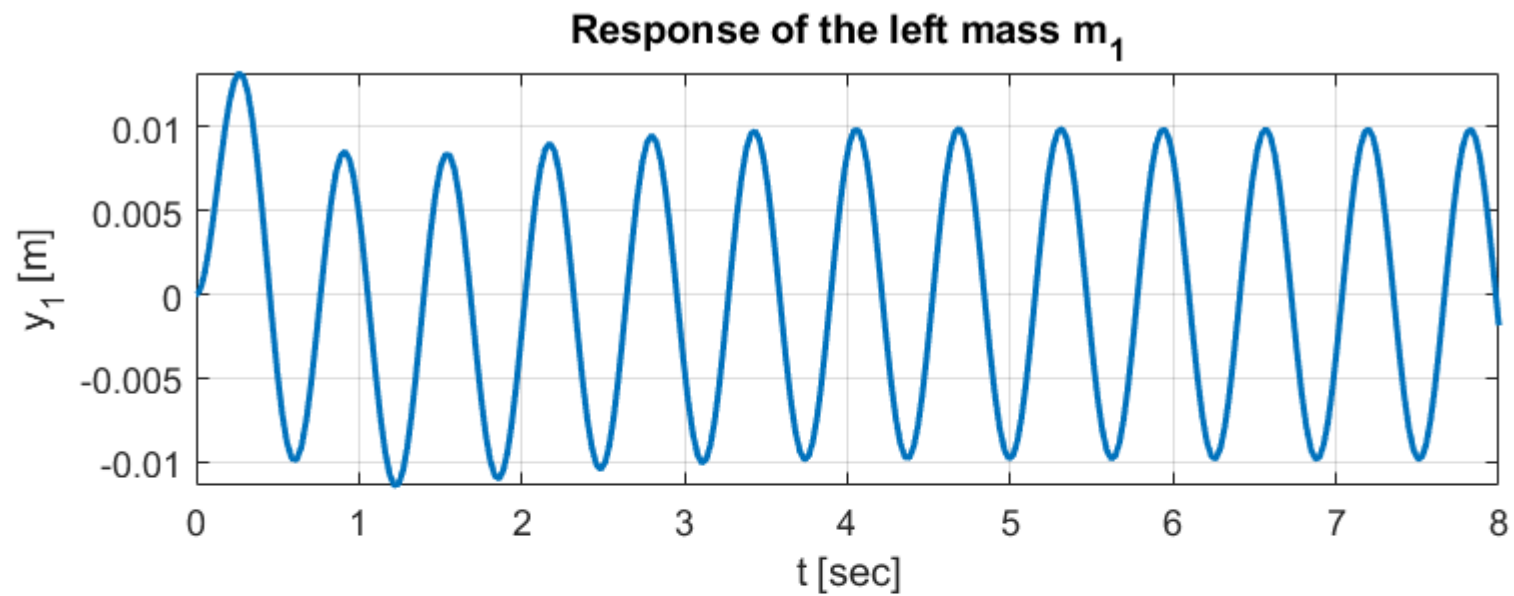
function wdot = dwdt(t,w)
% Formulation using w = [y1 y1dot y2 y2dot]
global m c k F0 omegaf
wdot(1,1) = w(2);
wdot(2,1) = (F0*cos(omegaf*t) - (3*c*w(2)-c*w(4)+2*k*w(1)-k*w(3)))/m;
wdot(3,1) = w(4);
wdot(4,1) = (-(-c*w(2)+3*c*w(4)-k*w(1)+2*k*w(3)))/m;
end

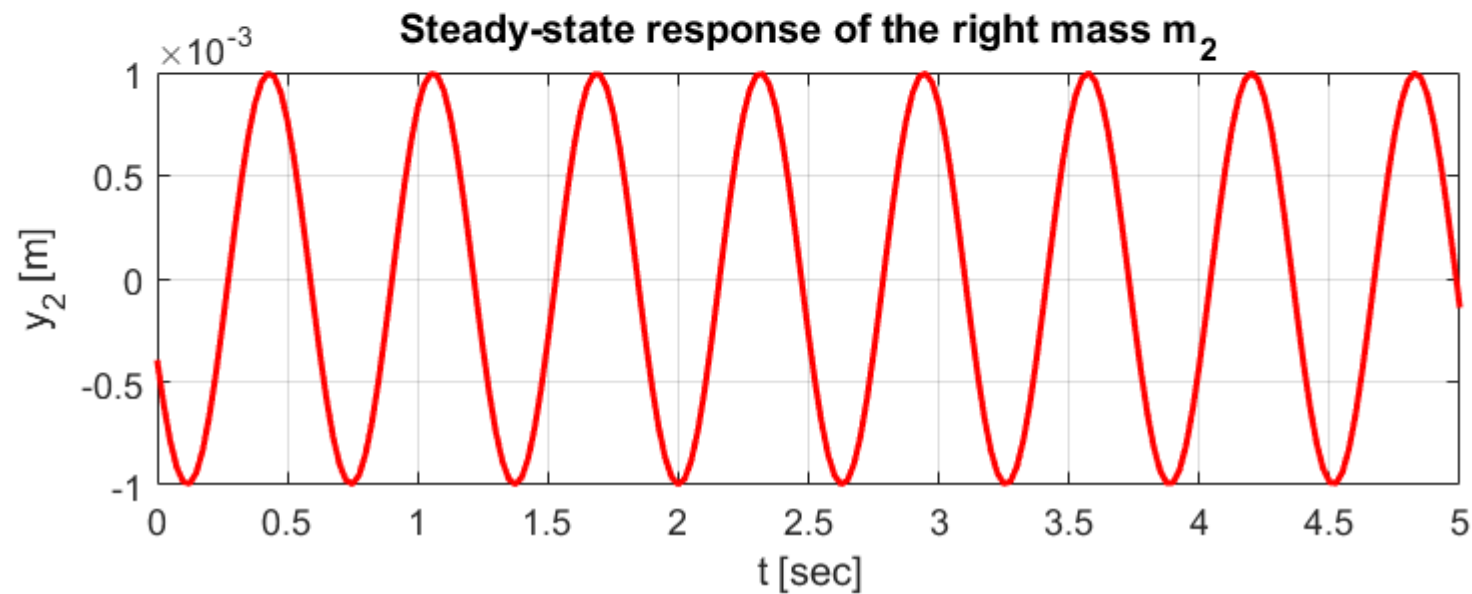
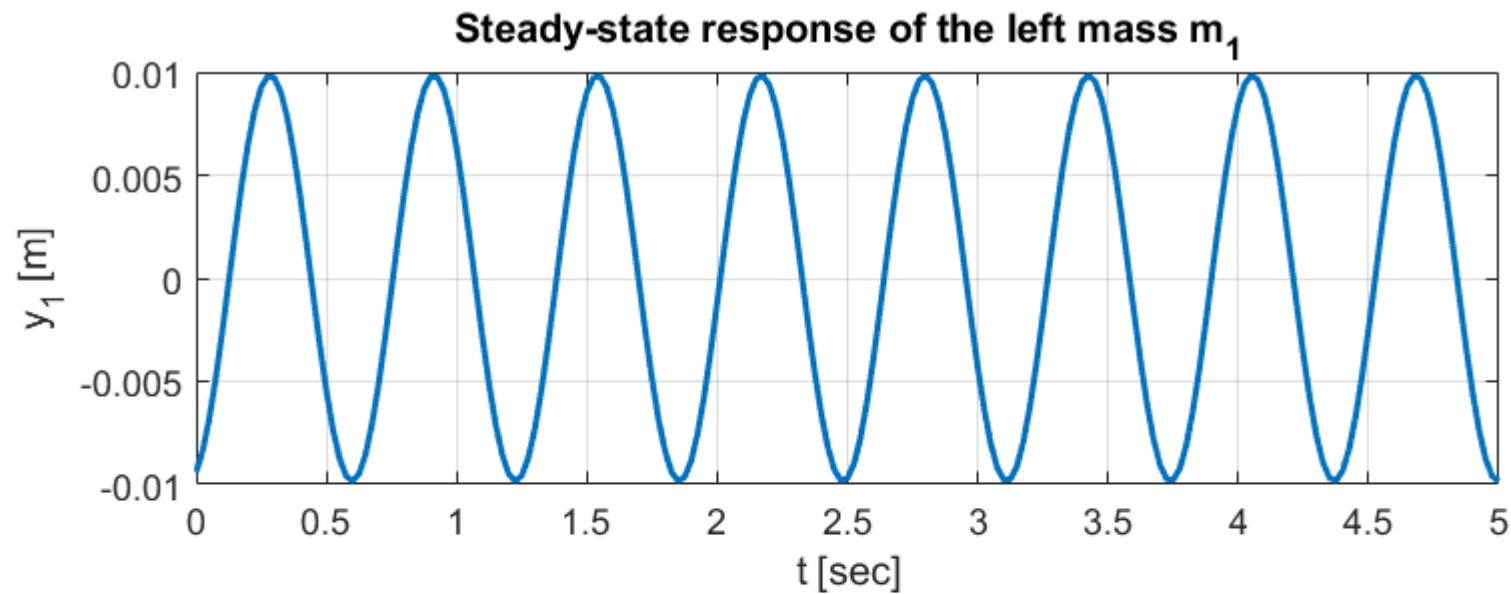
```

```

% function wdot = dwdt(t,w)
% % Matrix formulation using w = [y1 y2 y1dot y2dot]
% global invM C K F0 omegaf
% F = [F0*cos(omegaf*t); 0]; % external force vector
% wdot = [zeros(2) eye(2); -invM*K -invM*C]*w + [zeros(2,1); invM*F];
% end

```





Note that this time counts 0 as the start of the steady-state response.

Example: A 2DOF spring-mass-damper system

Determine and plot the time response of a two-degree-of-freedom system with equations of motion

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 4 & -1 \\ -1 & 2 \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} + \begin{bmatrix} 5 & -2 \\ -2 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} \cos 3t \quad (\text{E.1})$$

with the initial conditions

$$x_1(0) = 0.2, \quad \dot{x}_1(0) = 1.0, \quad x_2(0) = 0, \quad \dot{x}_2(0) = 0 \quad (\text{E.2})$$

First determine whether the damping is proportional or not. Is $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$?

$$\begin{bmatrix} 4 & -1 \\ -1 & 2 \end{bmatrix} = \alpha \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} + \beta \begin{bmatrix} 5 & -2 \\ -2 & 3 \end{bmatrix}$$

Since the off-diagonal terms of $\mathbf{M}$ are zero, so $\beta = 0.5$ is the only possibility. Then we have $\alpha + 5\beta = 4 \Rightarrow \alpha = 1.5$ and $2\alpha + 3\beta = 2 \Rightarrow \alpha = 0.25$. No possible solution. So damping is not proportional, and solution is obtained numerically.

Example: A 2DOF spring-mass-damper system

Note that proportional damping is only a sufficient condition for the governing equations to be decoupled by the modal vectors of the undamped system. In other words, $\mathbf{U}^T \mathbf{C} \mathbf{U}$ may still be diagonal even when the damping is not proportional, where $\mathbf{U}$ is the modal matrix of the undamped system. If $\mathbf{U}^T \mathbf{C} \mathbf{U}$ is not diagonal, then the system of equations cannot be decoupled and we would have to resort to numerical methods. Check whether $\mathbf{U}^T \mathbf{C} \mathbf{U}$ is diagonal or not. Consider the eigenvalue problem of the undamped system

$$\mathbf{K} \mathbf{u} = \lambda \mathbf{M} \mathbf{u}$$

We will solve using MATLAB eig

```
>> M = eye(2); K = [5 -2; -2 3];  
>> [V,D] = eig(K,M)
```

```
V =  
    -0.5257    -0.8507  
    -0.8507     0.5257
```

```
D =  
    1.7639     0  
     0     6.2361
```

```
>> V.'*M*V
```

```
    1.0000     0  
     0     1.0000
```

```
>> V.'*K*V
```

```
    1.7639   -0.0000  
   -0.0000    6.2361
```

```
>> V.'*C*V
```

```
    1.6584    0.4472  
    0.4472    4.3416
```

$\mathbf{V}^T \mathbf{C} \mathbf{V}$ is not diagonal,
hence eqns. of motion
cannot be decoupled

Example: A 2DOF spring-mass-damper system

Solution: In order to use the MATLAB program `ode23`, the two coupled second-order differential equations, Eq. (E.1), are to be expressed as a system of coupled first-order differential equations. For this, we introduce new variables y_1 , y_2 , y_3 , and y_4 as

$$y_1 = x_1, \quad y_2 = \dot{x}_1, \quad y_3 = x_2, \quad y_4 = \dot{x}_2$$

and express Eq. (E.1) as

$$\ddot{x}_1 + 4\dot{x}_1 - \dot{x}_2 + 5x_1 - 2x_2 = \cos 3t \quad (\text{E.3})$$

or

$$\dot{y}_2 = \cos 3t - 4y_2 + y_4 - 5y_1 + 2y_3 \quad (\text{E.4})$$

and

$$2\ddot{x}_2 - \dot{x}_1 + 2\dot{x}_2 - 2x_1 + 3x_2 = 2 \cos 3t \quad (\text{E.5})$$

or

$$\dot{y}_4 = \cos 3t + \frac{1}{2}y_2 - y_4 + y_1 - \frac{3}{2}y_3 \quad (\text{E.6})$$

Example: A 2DOF spring-mass-damper system

Thus Eq. (E.1) can be restated as

$$\begin{Bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \\ \dot{y}_4 \end{Bmatrix} = \begin{Bmatrix} y_2 \\ \cos 3t - 4y_2 + y_4 - 5y_1 + 2y_3 \\ y_4 \\ \cos 3t + \frac{1}{2}y_2 - y_4 + y_1 - \frac{3}{2}y_3 \end{Bmatrix} \quad (\text{E.7})$$

with the initial conditions

$$\vec{y}(0) = \begin{Bmatrix} y_1(0) \\ y_2(0) \\ y_3(0) \\ y_4(0) \end{Bmatrix} = \begin{Bmatrix} 0.2 \\ 1.0 \\ 0.0 \\ 0.0 \end{Bmatrix} \quad (\text{E.8})$$

Example: A 2DOF spring-mass-damper system

The MATLAB program to solve Eq. (E.7) with the initial conditions of Eq. (E.8) is given below.

```
% Ex5_20.m
tspan = [0: 0.01: 20];
y0 = [0.2; 1.0; 0.0; 0.0];
[t,y] = ode23('dfunc5_15', tspan, y0);
subplot (211)
plot (t,y (:, 1));
xlabel ('t');
ylabel ('x1 (t)');
subplot (212)

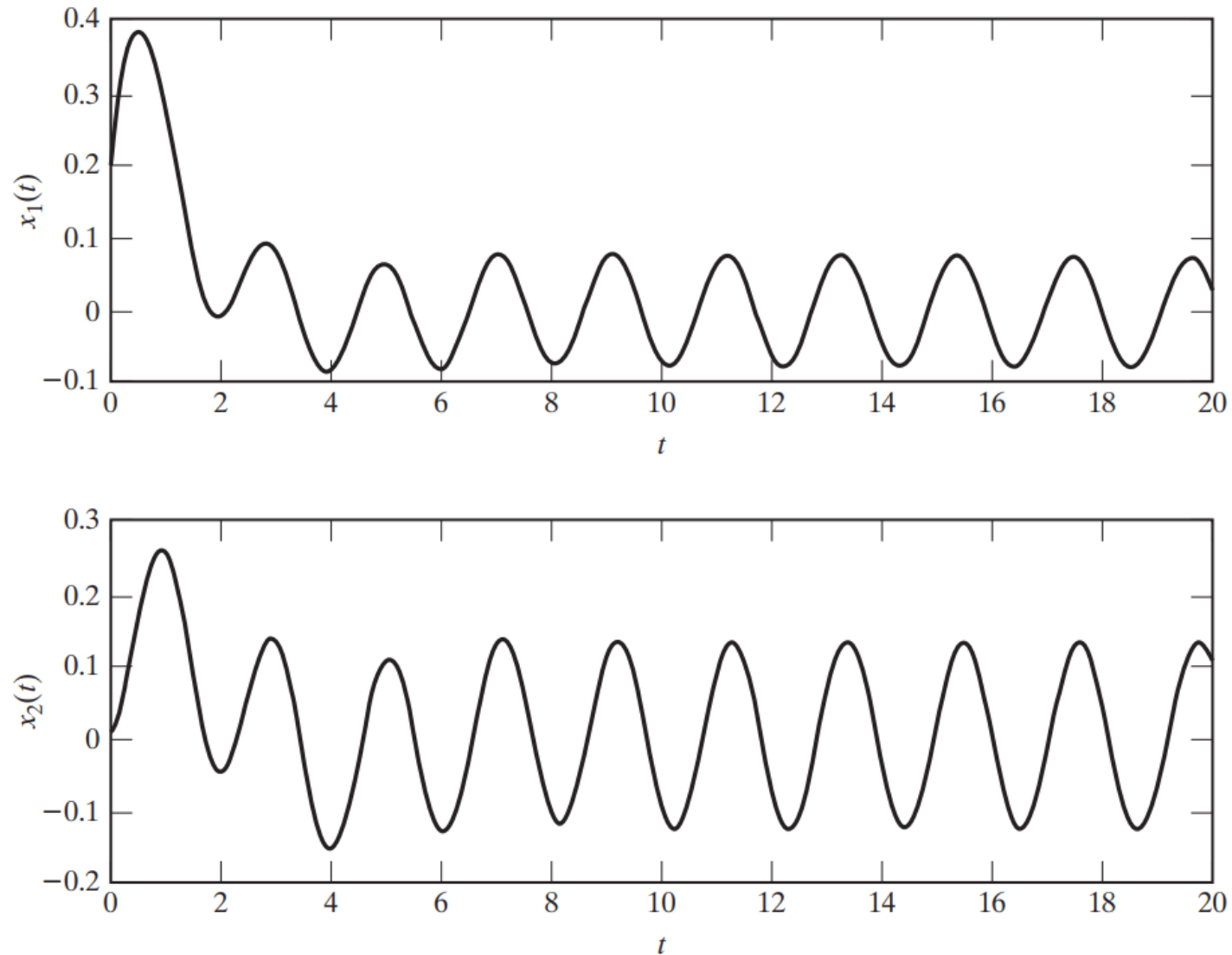
plot (t,y (:, 3));
xlabel ('t');
ylabel ('x2 (t)');

%dfunc5_15.m
function f = dfunc5_15(t,y)
f = zeros(4, 1);
f(1) = y(2);
f(2) = cos(3*t) - 4*y(2) + y(4) - 5*y(1) + 2*y(3);
f(3) = y(4);
f(4) = cos(3*t) + 0.5*y(2) - y(4) + y(1) - 1.5*y(3);
```

@dfunc5_15

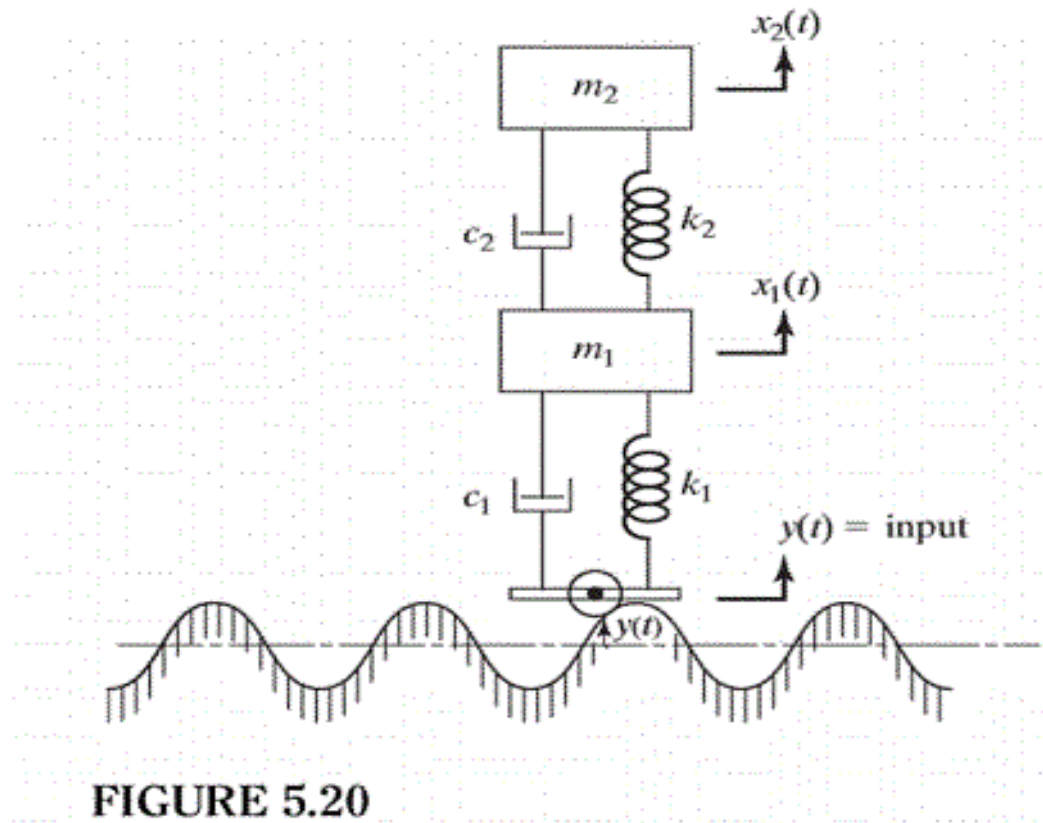
function m-file

Example: A 2DOF spring-mass-damper system



Example: A 2DOF suspension model traversing a road

Derive the equations of motion of the system shown in Fig. 5.20.



Solution

quarter car
model for
a vehicle

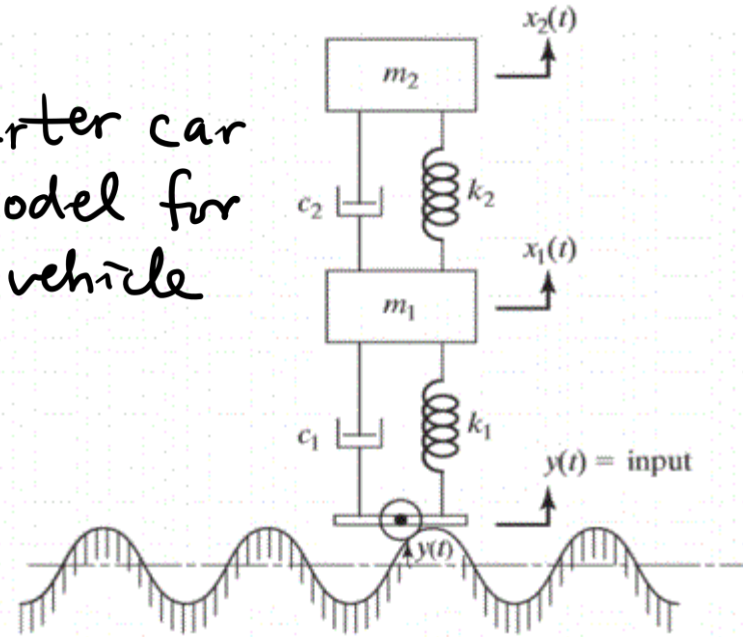
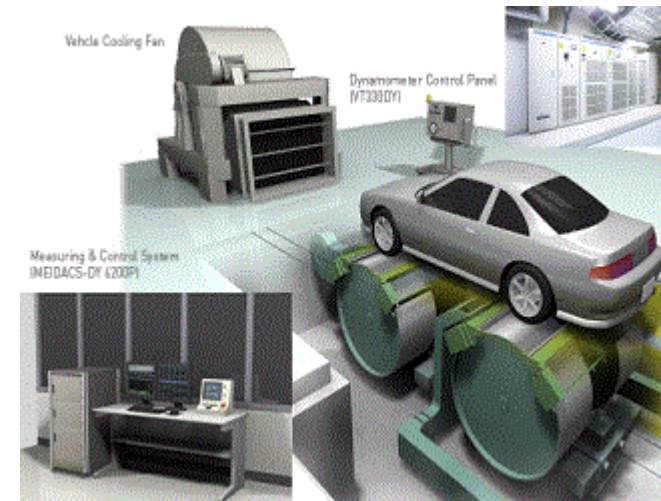


FIGURE 5.20



https://www.superflow.com/userdocs/products/SF-849_500x250.png



https://www.meidensha.com/products/automobile/prod_01/prod_01_02/prod_01_02_03/_icsFiles/artimage/2015/05/01/ce5_fb03/f-1-1-6-img-04.jpg

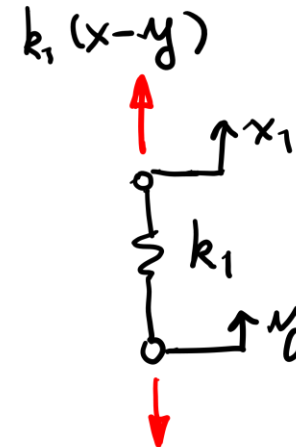
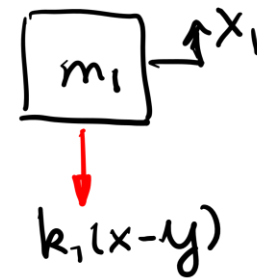
Solution / formulation of governing equations:

Egns of motion:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$= \begin{Bmatrix} k_1 y + c_1 \dot{y} \\ 0 \end{Bmatrix}$$

$$m_1 \ddot{x}_1 = \dots -k_1(x-y)$$



Example

Rao 5.50. A trailer of mass M , connected to a wall through a spring of stiffness k and a damper of damping coefficient c , slides on a frictionless surface, as shown in Fig. 5.49. A uniform rigid bar, pin-connected to the trailer, can oscillate about the hinge point O . Derive the equations of motion of the system under the applied forces $F(t)$ and $M_t(t)$ indicated in the figure.

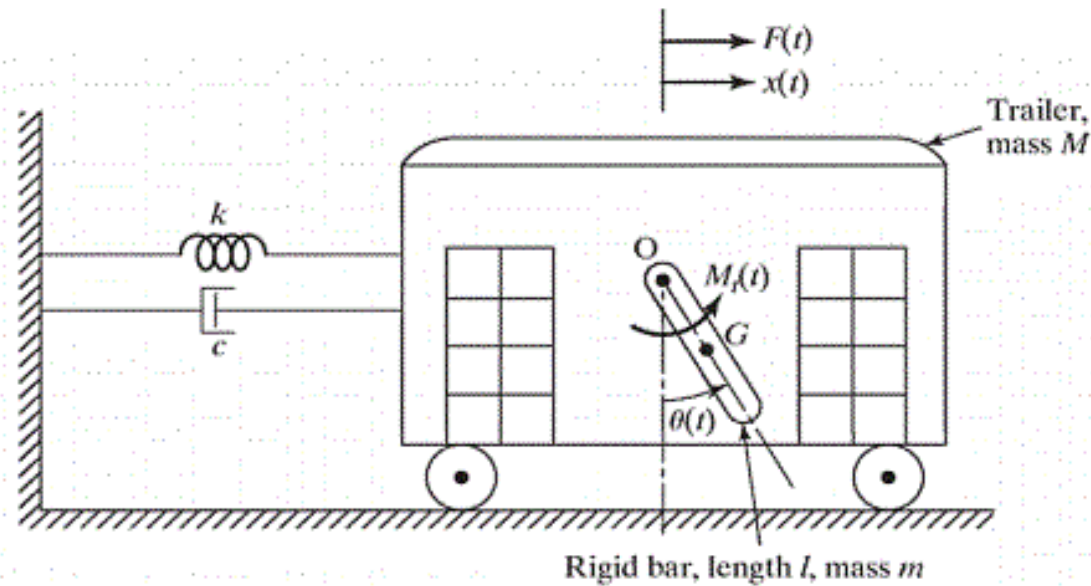


FIGURE 5.49

Solution

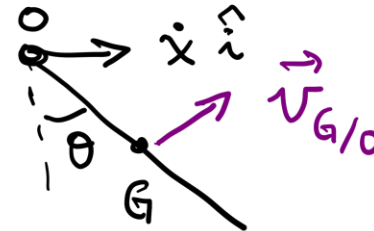
Applications: Use of IMU (inertial measurement unit) sensors in vehicles.
Derive equations of motion by Lagrange's method:

$$T = T_{\text{trailer}} + T_{\text{bar}}$$

$$T_{\text{trailer}} = \frac{1}{2} M \dot{x}^2$$

$$\begin{aligned} T_{\text{bar}} &= T_{\text{translational}} + T_{\text{rotational}} \\ &= \frac{1}{2} m v_G^2 + \frac{1}{2} (I_G) \dot{\theta}^2 \end{aligned}$$

G = c.m. of the bar



Solution

$$\vec{V}_G = \dot{x} \hat{i} + \frac{l}{2} \dot{\theta} \hat{e}_t$$

$$= \dot{x} \hat{i} + \frac{l}{2} \dot{\theta} (\cos \theta \hat{i} + \sin \theta \hat{j})$$

$$V_G^2 = (\dot{x} + \frac{l}{2} \dot{\theta} \cos \theta)^2 + (\frac{l}{2} \dot{\theta} \sin \theta)^2$$

$$= \dot{x}^2 + \frac{l^2}{4} \dot{\theta}^2 + l \dot{x} \dot{\theta} \cos \theta$$

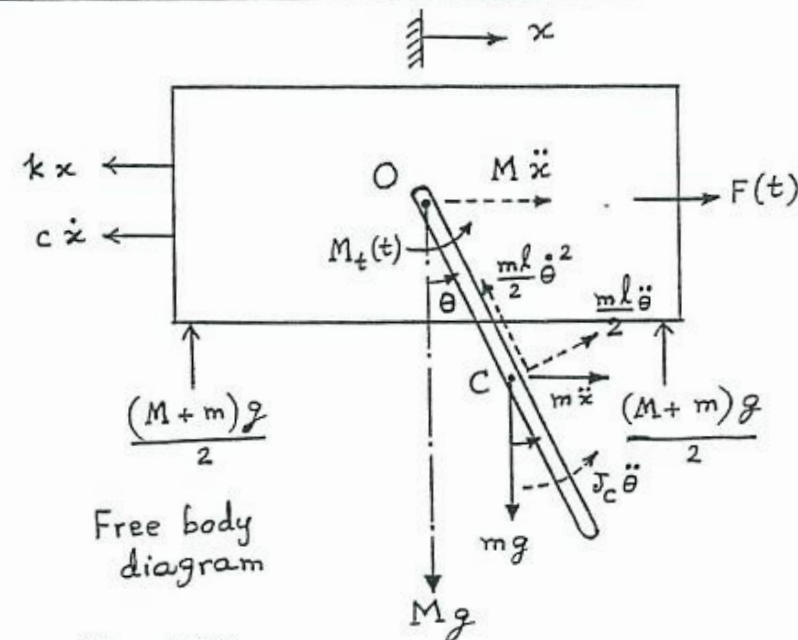
$$\Rightarrow T_{bar} = \frac{m}{2} (\dot{x}^2 + \frac{l^2}{4} \dot{\theta}^2 + l \dot{x} \dot{\theta} \cos \theta) + \frac{1}{2} \left(\frac{ml^2}{12} \right) \dot{\theta}^2$$

$$V_{spring} = \frac{1}{2} k x^2, \quad V_{bar} = -mg \frac{l}{2} \cos \theta$$

Apply Lagrange's eqn and then add damping to the x-eqn.

Rao 5.50

5.50



Equations of motion with coordinates $x(t)$ and $\theta(t)$:

For motion along x :

$$M \ddot{x} = -kx - c\dot{x} - (m\ell/2) \ddot{\theta} \cos \theta - m\ddot{x} + (m\ell/2) \dot{\theta}^2 \sin \theta + F(t) \quad (1)$$

For rotation about O :

$$J_c \ddot{\theta} + (m\ell/2) \ddot{\theta} (\ell/2) + m\ddot{x} (\ell \cos \theta/2) = -mg(\ell/2) \sin \theta + M_t(t) \quad (2)$$

Using $J_c = \frac{1}{12} m \ell^2$, $\cos \theta \approx 1$ and $\sin \theta \approx \theta$ and neglecting the nonlinear term involving $\dot{\theta}^2$ in Eq. (1), Eqs. (1) and (2) can be rewritten in matrix form as:

$$\begin{bmatrix} (M+m) & m\ell/2 \\ m\ell/2 & (J_c + m\ell^2/4) \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{\theta} \end{Bmatrix} + \begin{bmatrix} k & 0 \\ 0 & mg\ell/2 \end{bmatrix} \begin{Bmatrix} x \\ \theta \end{Bmatrix} = \begin{Bmatrix} F(t) \\ M_t(t) \end{Bmatrix}$$